

Day 12 - Notes

Salesforce Summer Program – Day 12 Notes

Goal for Today

Today's focus was understanding the professional Salesforce developer workflow used in real enterprise environments.

Main focus areas:

- Salesforce DX
 - Salesforce CLI
 - GitHub integration
 - Team collaboration
 - Source-driven development
 - Enterprise deployment workflow
 - Agentforce DX overview
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1. Salesforce DX (Developer Experience)

What is Salesforce DX?

Salesforce DX is a modern development model and toolset for Salesforce application development.

It supports:

- Source-driven development
- Team collaboration
- CLI-based workflows
- Version control integration
- Automated deployment processes

Salesforce DX changes development from browser-only customization to professional software engineering practices.

Why Salesforce DX is Important

Traditional Salesforce development mainly depended on browser clicks.

Problems with browser-only development:

- Difficult collaboration
- No proper version tracking
- Hard rollback process
- Manual deployment issues
- High risk of production errors

Salesforce DX solves these issues by enabling:

- Source code management
 - Structured project workflow
 - Easier deployment
 - Better testing
 - Team collaboration
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Source-Driven Development

In Salesforce DX:

- Source code becomes the main source of truth.
- Metadata is stored in files locally.
- Changes are tracked using GitHub.

Benefits:

- Easier recovery
 - Better collaboration
 - Safer deployments
 - Complete change tracking
 - Better scalability
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2. Salesforce CLI (Command Line Interface)

What is Salesforce CLI?

Salesforce CLI is a command-line tool used to interact with Salesforce orgs.

Developers use CLI to:

- Create projects
 - Deploy metadata
 - Retrieve metadata
 - Run tests
 - Open orgs
 - Authorize orgs
 - Manage environments
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Why CLI Matters

CLI improves productivity because developers can automate repetitive tasks.

Benefits of CLI:

- Faster workflow
 - Automation support
 - Better developer productivity
 - CI/CD integration
 - Easier deployment management
 - Reduced manual effort
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Common Salesforce CLI Commands

Authorize Org

```
sf org login web
```

Open Org

```
sf org open
```

Deploy Source

```
sf project deploy start
```

Retrieve Metadata

```
sf project retrieve start
```

View Org List

```
sf org list
```

3. GitHub and Version Control

Why GitHub Matters

GitHub helps enterprise teams manage source code safely and collaboratively.

GitHub provides:

- Source backup
 - Version history
 - Collaboration workflow
 - Branch management
 - Change tracking
 - Rollback capability
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Importance of Version Control

Without version control:

- Developers overwrite each other's work

- Bugs become difficult to trace
- Recovery becomes difficult
- Production deployments become risky

Version control ensures:

- Stable collaboration
 - Safer deployments
 - Better accountability
 - Organized workflow
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Branches in Enterprise Development

Enterprise teams use branches to isolate work.

Common branches:

- Main branch
- Development branch
- Feature branches
- Release branches
- Hotfix branches

Benefits:

- Prevents unstable code from affecting production
 - Allows parallel development
 - Supports controlled releases
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4. Team Collaboration Workflow

Problems Without Collaboration Tools

If multiple developers work on the same Salesforce project without proper workflow:

Common Problems

1. Code Conflicts

Different developers may modify the same files.

2. Overwritten Changes

One developer may accidentally replace another developer's work.

3. Deployment Failures

Untracked dependencies may break deployments.

4. Lack of Accountability

Teams cannot identify who made changes.

5. Production Instability

Direct production changes increase risk.

6. Difficult Rollback

Recovering previous stable versions becomes difficult.

Enterprise Collaboration Solutions

Enterprise teams use:

- GitHub repositories
- Branching strategies
- Pull requests
- CI/CD pipelines
- Sandboxes
- Automated testing
- Deployment validation

These processes improve:

- Stability
- Reliability
- Team productivity

- Deployment safety
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5. Org Development Model

Why Development and Production Must Be Separate

Enterprise systems never allow direct development in production.

Reasons:

- Prevents business disruption
 - Protects customer data
 - Supports testing
 - Improves deployment safety
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Salesforce Environment Types

Developer Sandbox

Used for individual development work.

Partial Sandbox

Used for integration testing.

Full Sandbox

Used for staging and regression testing.

Production Org

Live environment used by real users.

Benefits of Sandbox Workflow

- Safer development
- Easier testing

- Controlled deployment
 - Better rollback capability
 - Reduced production risk
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6. Enterprise Software Engineering

College Coding vs Enterprise Development

College Coding

Usually includes:

- Small projects
- Single developer
- Minimal testing
- Temporary usage
- Simple deployment

Enterprise Development

Requires:

- Large teams
 - Collaboration workflow
 - Structured testing
 - Deployment planning
 - Rollback capability
 - Reliability engineering
 - Long-term maintenance
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Enterprise Reliability Requirements

Enterprise applications must support:

- Thousands of users
- Continuous uptime

- Secure operations
 - Stable deployments
 - Large datasets
 - Recovery during failures
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7. Deployment Workflow

Why Deployment Workflow is Important

Deployment workflow controls how changes move safely between environments.

Benefits:

- Prevents production failures
 - Improves testing quality
 - Supports rollback
 - Reduces downtime
 - Maintains system stability
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Enterprise Deployment Lifecycle

Development

Developers create features locally or in sandboxes.

Version Control

Changes are committed to GitHub.

Testing

Automated and manual tests are performed.

Validation

Deployments are validated before release.

Staging

Changes are tested in staging environments.

Production Deployment

Final deployment happens after approval.

8. Agentforce DX Overview

What is Agentforce DX?

Agentforce DX extends Salesforce DX tools for AI agent development.

It supports:

- Agent development using source code
 - CLI-based workflows
 - Agent deployment
 - Validation and preview
 - Team collaboration
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Agentforce DX Features

Agent Script Files

Define agent behavior and logic.

Preview and Validation

Developers can test agents before deployment.

CLI Integration

Agents can be deployed using Salesforce CLI.

Source Control Integration

Agent metadata can be stored in GitHub.

9. Realizations from Day 12

Key Learnings

- Enterprise development is much more than writing code.
 - Real-world systems require collaboration and structured workflow.
 - GitHub is critical for version management.
 - CLI tools improve developer efficiency.
 - Deployment workflow protects production systems.
 - Rollback capability is essential for reliability.
 - Salesforce DX modernizes Salesforce development.
 - Enterprise engineering focuses heavily on stability and teamwork.
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10. Key Concepts Learned

- Salesforce DX supports source-driven development
 - Salesforce CLI improves automation and productivity
 - GitHub enables collaboration and version tracking
 - Enterprise systems require deployment planning
 - Teams need branching and workflow management
 - Sandboxes improve development safety
 - Rollback capability protects enterprise systems
 - Large-scale projects require structured engineering workflow
 - Agentforce DX extends Salesforce DX into AI development
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Reflection

Learning Salesforce DX and enterprise workflows showed how professional software engineering differs from simple coding. Real enterprise systems require collaboration, version control, deployment management, testing, rollback capability, and structured workflows.

Modern development is not only about creating features but also about maintaining reliability, scalability, and stability while multiple developers work together on the same system.

Salesforce DX, CLI tools, GitHub, and deployment workflows help enterprise teams build safer, scalable, and production-ready applications efficiently.

Final Outcome

Developed understanding of professional Salesforce development workflow, Salesforce DX, Salesforce CLI, GitHub collaboration, enterprise deployment lifecycle, sandbox strategy, source-driven development, and Agentforce DX concepts used in real-world enterprise engineering environments.